

Rohit Kumar

B.Tech Computer Science & Applied Mathematics | GGSIPU, New Delhi

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Summary

B.Tech Computer Science student with hands-on experience in full-stack development and DevOps. Skilled in building and deploying scalable applications using MERN and Spring Boot, with experience in CI/CD pipelines, container orchestration (Docker, Kubernetes), and Infrastructure as Code (Terraform). Developed 3+ production-style projects with automated deployment, monitoring, and security integration.

Education

Bachelor of Technology (B.Tech) – Computer Science & Applied Mathematics

2024 – 2028

Vivekananda Institute of Professional Studies - Technical Campus, New Delhi

CGPA: 7.7/10

Technical Experience

- Built React frontends using React Router, Axios, and Tailwind CSS with JWT-based authentication and role-based access control.
- Developed full-stack applications using Node.js, Express, and MongoDB with RESTful APIs and multi-role user systems.
- Built and containerized Spring Boot backend services with MySQL/PostgreSQL using Docker.
- Used Linux for shell scripting, process management, permissions, and application deployment.
- Designed Jenkins CI/CD pipelines with SonarQube, OWASP, Docker, Trivy, and automated deployment stages.
- Worked with Kubernetes (pods, services, deployments, ConfigMaps) and Helm for container orchestration.
- Implemented monitoring using Prometheus and Grafana for application and system metrics.
- Automated infrastructure provisioning and configuration using Terraform and Ansible (IaC).

Projects

LibraFlow - Library Management System

GitHub

React | Node.js | MongoDB | Cloudinary | Jenkins | Linux

2025

- Developed a full-stack library system with **3 user roles** and **15+ REST API endpoints** for book management, borrowing workflows, and reporting.
- Integrated Cloudinary for media storage, reducing backend load and improving response time for image handling.
- Implemented Jenkins-based CI pipelines, reducing manual build time from **10 minutes to <2 minutes** and enabling faster development cycles.
- Automated fine calculation and QR-based shelf identification, improving operational efficiency.

3-Tier Java App with Docker Orchestration

GitHub

Java | Spring Boot | MySQL | Docker | Kubernetes | Maven | Terraform | Jenkins | AWS EC2

2026

- Built a **3-tier application** (UI, backend, DB) using Spring Boot and MySQL; containerized services with Docker and orchestrated using Docker Compose.
- Provisioned **dev, test, and prod environments** on AWS EC2 using Terraform, reducing infrastructure setup time from **1 hour to <10 minutes**.
- Implemented a **CI/CD pipeline** using Jenkins (build, test, deploy), enabling automated multi-environment deployments in **<5 minutes**.
- Developed Kubernetes manifests to enable container orchestration, improving scalability and deployment consistency.

Load Balancing Simulator

GitHub

React.js | Node.js | Express.js | PostgreSQL | Docker | Kubernetes | Jenkins | Prometheus | Grafana

2026

- Built a distributed systems simulator implementing **3 load balancing algorithms** - Round Robin, Weighted Round Robin, and Least Connections - to analyze traffic flow, queue buildup, and fault tolerance.
- Developed a custom **discrete-event simulation engine** generating real-time metrics for **P95 latency, throughput, and server utilization** with interactive dashboards.
- Automated CI/CD using **Jenkins, SonarQube, and Trivy**, reducing deployment and security validation time from **35 minutes to <5 minutes**.
- Containerized services with Docker and deployed on Kubernetes using **Prometheus + Grafana monitoring**, reducing environment setup time from **25 minutes to <4 minutes**.

Technical Skills

Languages: JavaScript, Java, C, Bash, YAML

Frontend: React, Tailwind CSS, Vite

Backend: Node.js, Express, Spring Boot (REST APIs)

DevOps & Cloud: Docker, Kubernetes, Helm, Jenkins, AWS (EC2, S3, RDS, DynamoDB)

CI/CD & IaC: Jenkins, Terraform, Ansible, Git

Databases: MongoDB, MySQL, PostgreSQL

Monitoring: Prometheus, Grafana

Security: SonarQube, Trivy, OWASP, JWT, OAuth, RBAC

Tools: Linux, Docker Hub, Cloudinary, Postman, VS Code, Vim